

TREMORTRACE · DESIGN PRECIS

TremorTrace design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/tremortrace		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

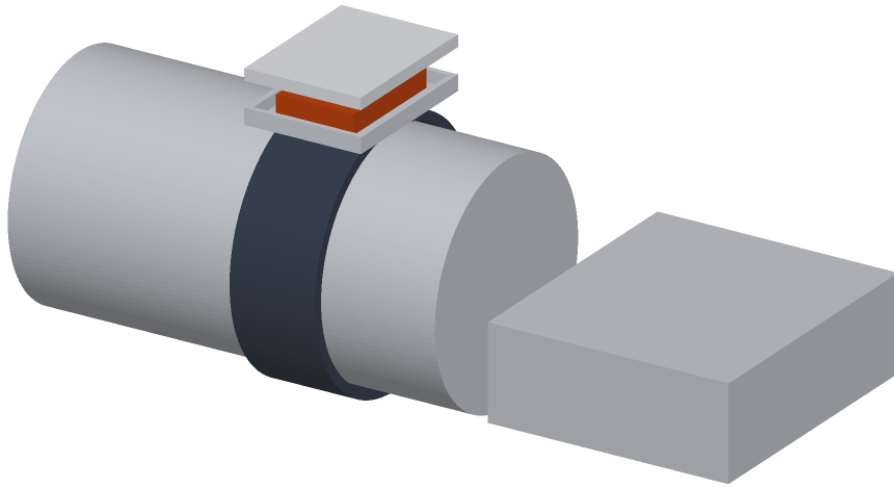
Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

TremorTrace design precis

TremorTrace is a small wrist pod on a watch strap that records wrist motion all day, turns it into 10-second tremor summaries on the band, and hands them to an open notebook that draws a daily tremor profile. First-order numbers suggest a single off-the-shelf module and a 150 mAh cell can meet every requirement for about \$40 in parts.

TremorTrace

CONCEPT, NOT FOR FABRICATION



Grey: Forearm and hand for scale

■ How it works

1. **Sense.** A 6-axis IMU (accelerometer and gyroscope) samples wrist motion at 104 Hz.
2. **Summarize on the band.** Every 10 seconds the firmware band-pass filters the signal to 3 to 15 Hz, runs a spectrum, and stores a compact summary: dominant frequency, band power, acceleration and angular velocity RMS, and an activity flag so voluntary movement is not counted as tremor.
3. **Store.** Summaries go to on-board flash, so the band works for a week without a phone.
4. **Sync.** When the owner opens the phone app or a laptop script, summaries transfer over Bluetooth Low Energy. Nothing goes to the cloud.
5. **Analyze.** An open Python notebook turns summaries into a daily profile: minutes of tremor per hour, dominant frequency and amplitude over the day, and optional medication-time markers entered by the owner.

TremorTrace: data flow (local-first, no cloud)

CONCEPT, NOT FOR FABRICATION



■ Main components

#	Component	Proposed choice	Notes
1	Strap	22 mm silicone watch strap	Standard spring bars; skin-safe
2	Enclosure lid	3D-printed, PETG or resin	Light pipe for the status LED
3	Controller and IMU	nRF52840 module with built-in 6-axis IMU, 2 MB flash and LiPo charger (Seeed XIAO nRF52840 Sense class)	One module, no custom PCB. Proposed, awaiting Amish
4	Battery	150 mAh protected LiPo	See power budget
5	Enclosure base	3D-printed with TPU gasket	Charging contacts through the base
6	Charging contacts	Magnetic 2-pin pogo connector	Keeps the enclosure sealed

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3.

Quantity	Estimate	Basis	Requirement
Frequency resolution	0.1 Hz	10 s window at 104 Hz	R1 (0.2 Hz) met
Nyquist limit	52 Hz	104 Hz sampling	R1, R3 met
Average current while recording	about 1.5 mA	IMU about 0.9 mA, MCU processing about 0.5 mA, BLE about 0.1 mA	
Daily charge used	about 25 mAh	16 h recording plus 8 h at about 0.05 mA	
Battery life	about 4 days	150 mAh cell, 70 % usable	R4 (3 days) met, margin thin
Summary storage	about 185 kB per day	32 bytes per 10 s window, 16 h	
On-band history	about 10 days	2 MB flash, minus firmware space	R5 (7 days) met
Mass	about 22 g	Module 3 g, cell 3.5 g, enclosure 5 g, strap 10 g	R6 (30 g) met
Pod size	40 x 30 x 12 mm	Module 21 x 18 mm over a 30 x 20 x 5 mm cell	R6 met
Parts cost	about \$40	Indicative prices, see bom/bom.csv	R10 met

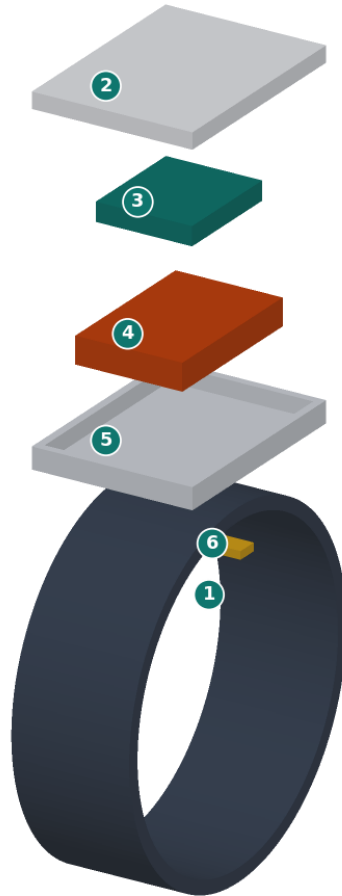
■ Key design choices

- **Summaries, not raw data.** Storing 10-second summaries instead of raw samples cuts storage by about 1,000 times and keeps personal data minimal. Short raw snippets can be captured on demand for research. Proposed, awaiting Amish.
- **One module, no custom PCB.** Keeps the first build within reach of anyone with a soldering iron and a 3D printer.
- **Local-first data.** No cloud account is needed to use or analyze the data (R9).

TremorTrace: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Silicone strap
- 2 Enclosure lid
- 3 Controller and IMU module
- 4 LiPo cell, 150 mAh
- 5 Enclosure base
- 6 Charging contacts



■ Safety

- Research and educational prototype only. It is not a medical device, has not been cleared or approved by any regulator, and must not be used to diagnose, treat or adjust medication.
- The LiPo cell sits against the skin: use a protected cell, never charge while worn, and stop use if the pod becomes warm or swollen.
- Use skin-safe strap materials and check for irritation during long wear.
- Tremor data is health data. Keep it on the owner's devices and share only with consent.

■ Open questions for TRL 3

- Validate the activity flag: how well can voluntary movement be separated from tremor with 10 s summaries?
- Confirm IMU current at 104 Hz in the chosen module and firmware.
- Choose phone app route: a simple web Bluetooth page or a native app. Proposed: web Bluetooth page, awaiting Amish.
- Identify a clinical or patient partner for co-design and eventual validation.

Concept media: [blueprint sheet](#), [interactive 3D model](#).